10/3/26 Exp. No:- 9.

Determination of z Parameters of Two port Network:-

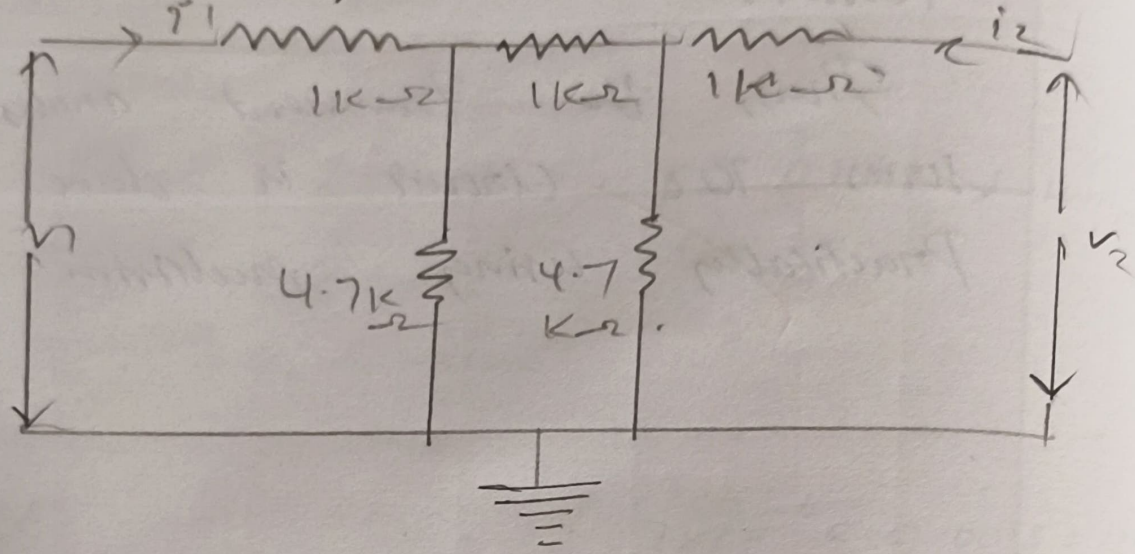
Aim:-

To calculate and verify z parameters of a two port network using Multisim simulator.

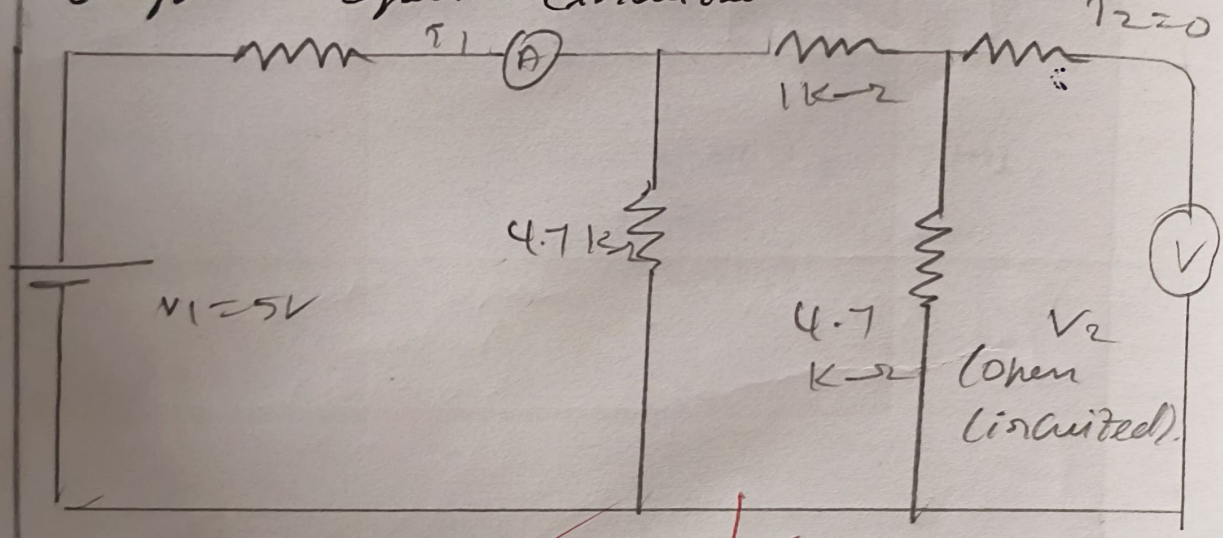
Apparatus Required:-

S.No	Components	Specification	Qnty.
1	Power supply	-	1
2	Multimeter	-	1
3	Resistors	10kΩ, 10kΩ, 4.7kΩ	
4	Bread board	-	1
5	Connecting wires	-	-
6	PC with multisim simulator	-	-

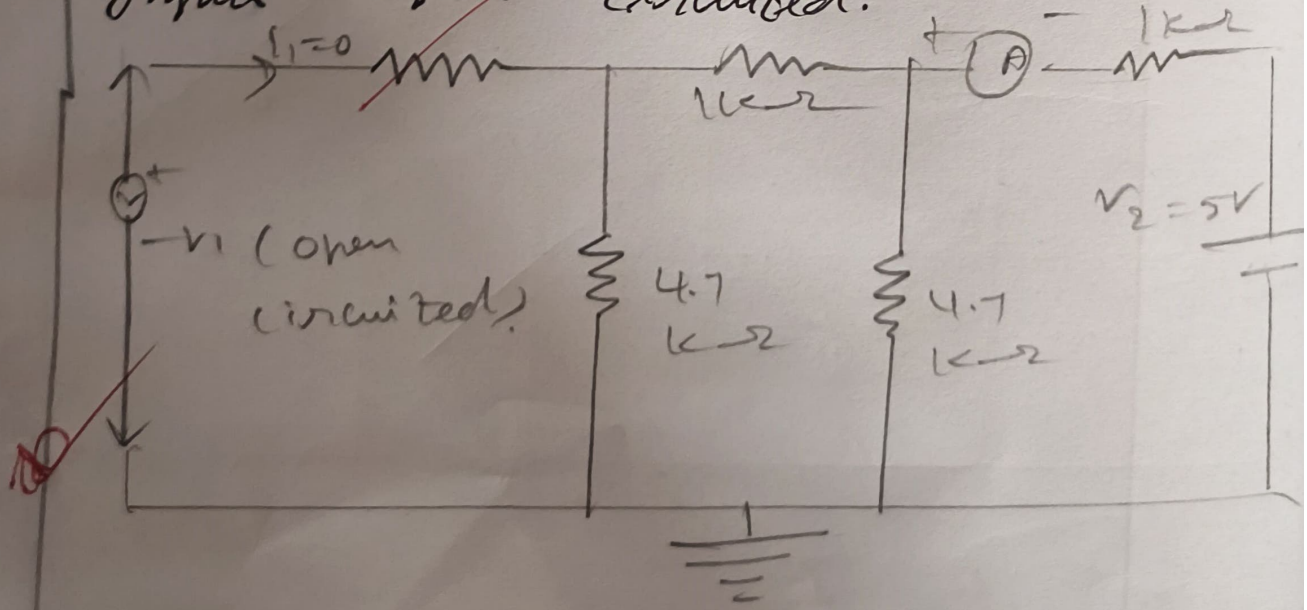
Circuit diagram:-



Output open circuited:



Input open circuited.

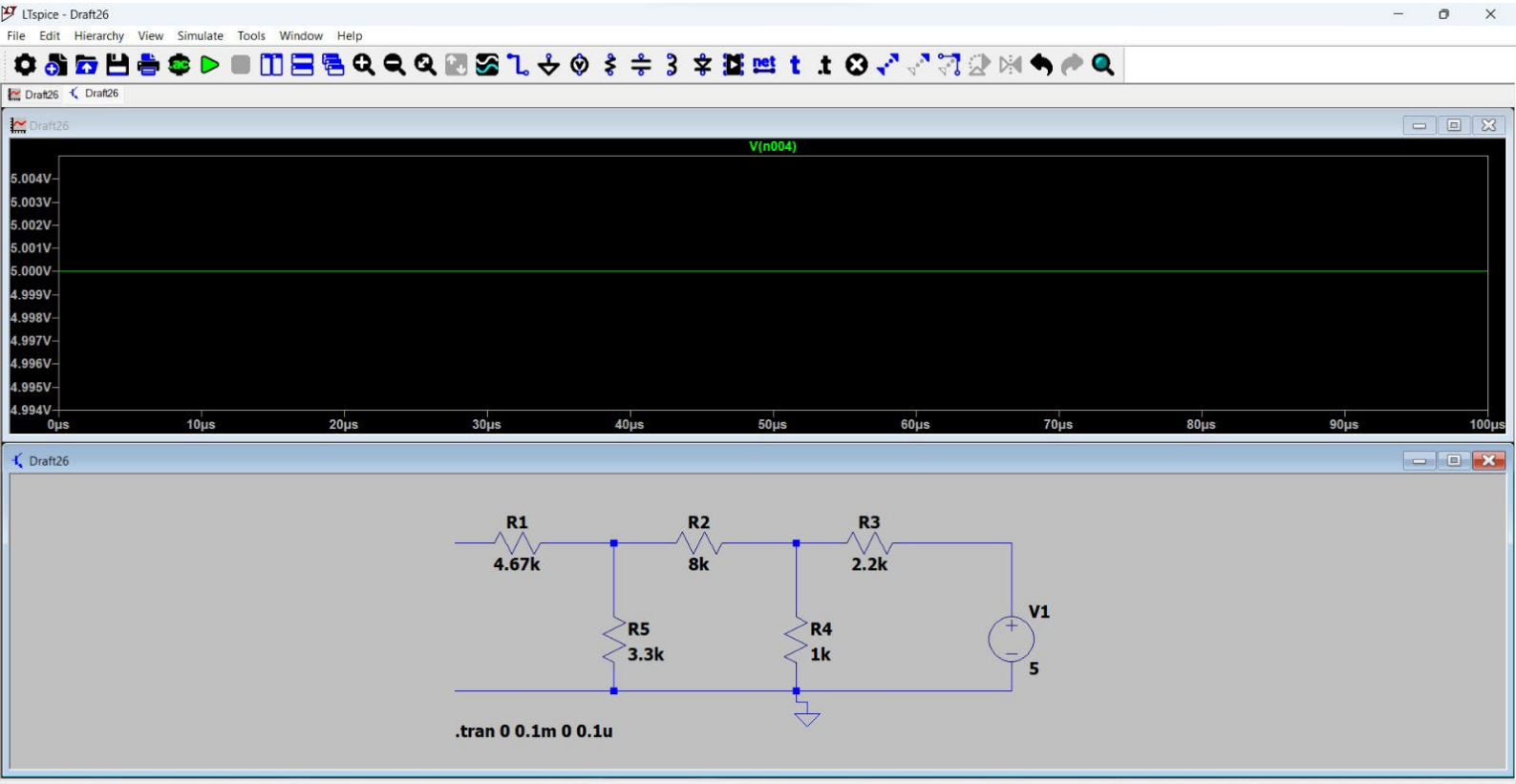


Observation:-

S.NO	output open circuited ($I_2=0$)			$Z_{21}=\frac{V_2}{I_1}(\Omega)$	$Z_{11}=\frac{V_1}{I_1}(\Omega)$
	V_1	V_2	I_1		
1)	5	2	2	$Z_{21}=\frac{2}{mA}$ $=1K\Omega$	$Z_{11}=\frac{5}{2mA}$ $=2.5K\Omega$
2)	6	3	3	$Z_{21}=\frac{3}{3mA}$ $=1K\Omega$	$Z_{11}=\frac{6}{3}$ $=3K\Omega$.

S.NO	Input open circuited ($I_1=0$)			$Z_{12}=\frac{V_1}{I_2}(\Omega)$	$Z_{22}=\frac{V}{I_2}(\Omega)$
	V_2	V_1	I_2		
1)	5	2	2	$Z_{12}=\frac{5}{2}$ $=2.5K\Omega$	$Z_{22}=\frac{5}{2}=2.5$
2)	6	3	3	$Z_{12}=\frac{3}{3}$ $=1K\Omega$	$Z_{22}=\frac{6}{3}$ $=2K\Omega$

all verified


LTspice - Draft26
File Edit Hierarchy View Simulate Tools Window Help
Draft26
V(n004)
5.004V
5.003V
5.002V
5.001V
5.000V
4.999V
4.998V
4.997V
4.996V
4.995V
4.994V
0µs
10µs
20µs
30µs
40µs
50µs
60µs
70µs
80µs
90µs
100µs
R1
4.67k
R2
8k
R3
2.2k
R5
3.3k
R4
1k
V1
5
.tran 0 0.1m 0 0.1u
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Procedure:-

a) Connect the circuit as shown in fig: and switch 'ON' the experimental board.

b). First open the O/P terminal and supply 5V to I/P terminal. measure O/P voltage and I/P current

c). Secondly, open I/P terminal and supply 5V to O/P terminal. measure I/P voltage and O/P current using multi-meter.

d). calculate the values of Z Parameter using equation 1 and 2.

e) Switch 'OFF' the supply after taking the readings.

Theoretical verification:-

$$Z_{11} = \frac{V_1}{I_1}\Big|_{I_2=0} = \frac{5}{2mA} = 2.5\ k\Omega$$

$$Z_{21} = \frac{V_2}{I_1}\Big|_{I_2=0} = \frac{2}{2mA} = 1k\Omega$$

$$Z_{12} = \frac{V_1}{I_2}\Big|_{I_1=0} = \frac{2}{2mA} = 1k\Omega$$

$$Z_{22} = \frac{V_2}{I_2}\Big|_{I_1=0} = \frac{5}{2mA} = 2.5k\Omega$$

$$Z = \begin{bmatrix} 2.5k\Omega & 1k\Omega \\ 1k\Omega & 2.5k\Omega \end{bmatrix}$$

Marks split-up:-

Sl. no	Description	Marks allotted	Marks obtained.
1	Experimental / setup / connection	20 marks	20
2	Observation / Readings	20 marks	10
3	Calculations / Results	20 marks	20
4	Viva	20 marks	10
5	Record	20 marks	20
		Total.	80

10/10/15

Result:-

The z- parameter of the two port network has been calculated and verified by using Multisim simulator.